

Show understanding of the need for input, output, primary memory and secondary (including removable) storage

9608_w21_qp_11_sc_4.a

QUESTION

- 4 Oliver needs to increase the secondary storage on his computer. He already has several hard disks.
- (a) Identify **two other** secondary storage devices that Oliver could use. Each device must be different.

1

2 [2]

MARK SCHEME

4(a)

One mark per bullet point to max 2

2

E.g.

- Optical Disc Drive // CD/DVD Drive/Writer
- Solid State Drive // USB Flash drive

9608_s20_qp_11_sc_2.a

QUESTION

- 2 Kal teaches Computer Science and uses different devices when teaching his students.
- (a) Tick (✓) **one or more** boxes on each row to indicate whether each device is an input device, an output device, or both.

Device	Input	Output
LCD monitor		
Microphone		
Keyboard		
Touchscreen		

[2]

MARK SCHEME

2(a)	1 mark for first three rows, 1 mark for the touchscreen being both.	2															
<table border="1"> <thead> <tr> <th>Device</th><th>Input</th><th>Output</th></tr> </thead> <tbody> <tr> <td>LCD Monitor</td><td></td><td>✓</td></tr> <tr> <td>Microphone</td><td>✓</td><td></td></tr> <tr> <td>Keyboard</td><td>✓</td><td></td></tr> <tr> <td>Touchscreen</td><td>✓</td><td>✓</td></tr> </tbody> </table>			Device	Input	Output	LCD Monitor		✓	Microphone	✓		Keyboard	✓		Touchscreen	✓	✓
Device	Input	Output															
LCD Monitor		✓															
Microphone	✓																
Keyboard	✓																
Touchscreen	✓	✓															

9608_w19_qp_13_sc_1.a.i

QUESTION

- 1 In a supermarket, a self-checkout machine allows customers to scan the barcodes of products and then pay for their shopping. These are an alternative to the traditional cashier-staffed checkout.



- (a) The self-checkout machine has a touchscreen.
- (i) Identify **two** other input devices that self-checkout machines have.

1

2

[2]

MARK SCHEME

1(a)(i)	1 mark per input device to max 2 e.g. Barcode scanner / Infra-red scanner Pressure sensor RFID / chip reader Bank note scanner Pin / key pad Magnetic strip reader	2
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9608_w19_qp_13_sc_1.a.ii

QUESTION

(ii) Identify **two** other output devices that self-checkout machines have.

1

2 [2]

MARK SCHEME

1(a)(ii)	1 mark per output device to max 2 e.g. Speaker Printer LCD screen	2
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9608_w19_qp_13_sc_1.b.i

QUESTION

(b) The self-checkout machines have primary storage.

(i) Give **two** reasons why the self-checkout machine needs primary storage.

1

.....

2

..... [2]

MARK SCHEME

1(b)(i)	1 mark per bullet point to max 2 To store the files needed to boot the system To store parts of the self-checkout machine operating system To store the self-checkout machine software To store the intermediate data / running total for items purchased	2
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9608_w19_qp_12_sc_6.a.i

QUESTION

- 6 Dominic uses a tablet computer to complete work. He records videos of his work to send to his colleagues to watch at a later date.

(a) The tablet computer has input and output devices.

(i) The table lists four devices built into the tablet.

Tick (✓) one or more boxes for each device to identify whether it is an input device, an output device or both.

Device	Input	Output
Touchscreen		
Webcam		
Microphone		
Fingerprint scanner		

[2]

MARK SCHEME

6(a)(i)	1 mark for touchscreen being both 1 mark for remaining 3 devices <table> <tr> <th>Device</th><th>Input</th><th>Output</th></tr> <tr> <td>Touchscreen</td><td>✓</td><td>✓</td></tr> <tr> <td>Webcam</td><td>✓</td><td></td></tr> <tr> <td>Microphone</td><td>✓</td><td></td></tr> <tr> <td>Fingerprint scanner</td><td>✓</td><td></td></tr> </table>	Device	Input	Output	Touchscreen	✓	✓	Webcam	✓		Microphone	✓		Fingerprint scanner	✓		2
Device	Input	Output															
Touchscreen	✓	✓															
Webcam	✓																
Microphone	✓																
Fingerprint scanner	✓																

9608_w19_qp_12_sc_6.b.i

QUESTION

(b) The tablet computer's secondary storage is solid state (flash) memory.

(i) Give **one** reason why the tablet computer needs secondary storage.

.....
..... [1]

MARK SCHEME

6(b)(i)	To store files / software long term	1
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9608_w19_qp_11_sc_2.a.i

QUESTION

2 Aaron uses a desktop computer to do school work.

(a) Aaron has a mouse and keyboard that he can use as input devices and a monitor as an output device.

(i) Identify **two** additional input devices Aaron could use with his desktop computer.

1
2 [2]

MARK SCHEME

2(a)(i)	1 mark per device to max 2 e.g. Trackpad/touchpad microphone touchscreen scanner	2
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9608_w19_qp_11_sc_2.a.ii

QUESTION

(ii) Identify **two** additional output devices Aaron could use with his desktop computer.

1
2 [2]

MARK SCHEME

2(a)(ii)	1 mark per device to max 2 e.g. printer speakers touchscreen	2
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9608_w19_qp_11_sc_2.a.iii

QUESTION

(iii) Aaron needs to store a large number of applications and data on his computer. He needs at least 50GB of secondary storage space.

Identify **one** internal secondary storage device for Aaron's computer.

.....

..... [1]

MARK SCHEME

2(a)(iii)	Magnetic hard disk drive // solid state drive	1
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9608_s18_qp_13_sc_7.b.i

QUESTION

(b) (i) Give **one** output device, other than a touchscreen, that may be part of this computer system.

.....[1]

MARK SCHEME

7(b)(i)	1 mark for suitable device e.g. Speaker Headphones	1
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QUESTION

- (ii) Give **two** input devices, other than a touchscreen, that may be part of this computer system. State how each device will be used by the visitors.

Device 1

Use

.....

Device 2

Use

.....

[4]

MARK SCHEME

7(b)(ii)	1 mark for naming input device 1 mark for use in this scenario, for max 2 devices e.g. Microphone ...visitor says commands / search criteria for the computer to respond to Keyboard ...visitor types key words to look for Mouse ...visitor controls cursor to navigate / select Trackpad ...visitor uses finger to control cursor to navigate / select	4
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QUESTION

(c) Give the most appropriate secondary storage device for this computer system.

Describe **two** reasons for your choice.

Device

Reason 1

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.....

.....

Reason 2

.....

.....

.....

[5]

MARK SCHEME

7(c)	1 mark for device (Internal) hard <u>drive</u> / solid state <u>drive</u> 1 mark per bullet. Max 2 marks for each reason, max two reasons e.g. Hard drive - Large capacity... to store videos / images / sound files with large file sizes - Reasonably fast access speed... Users will not have to wait for videos to load - Inexpensive per unit storage... If a large number of needed for different exhibits, the cost can be kept low - Does not need to be moved ... So moving parts unlikely to be damaged - Slower degradation of data ... So will last longer / be more reliable under heavy use Solid state - Large capacity... To store videos/images/sound files with large file sizes - Fast access speed... Users will not have to wait for videos to load - Reliable... Can be dropped/damaged and will likely still work / no moving parts - Quiet... No moving parts	5
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9608_s18_qp_11_sc_7.a

QUESTION

7 A student plays computer games on a games console.

(a) Identify **two** input devices and **one** output device used in a games console.

Input device 1

Input device 2

Output device

[3]

MARK SCHEME

7(a)	1 mark for each input device to max 2 e.g. (Handheld) remote controller Joystick / Games pad / joypad Accelerometer Microphone Suitable sensor 1 mark for output e.g. Motor/vibrator in joystick Speaker Screen/monitor /TV	3
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9608_w17_qp_13_sc_5.b.i

QUESTION

(b) (i) The user is considering the purchase of a removable device for secondary storage.

Name **one** suitable device.

.....[1]

MARK SCHEME

5(b)(i)	One mark from: External hard disk drive // SSD External CD / DVD drive Pen drive Blu-ray drive	1
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9608_w17_qp_13_sc_5.b.ii

QUESTION

(ii) Describe **two** possible uses for this device on a home Personal Computer (PC).

1

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2

.....

[2]

MARK SCHEME

5(b)(ii)	Two marks from: Additional secondary file storage // storing files Backup of files Archiving of files Transfer files to second computer	Max 2
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9608_w17_qp_12_sc_6.a

QUESTION

- 6 (a) A personal computer (PC) is extensively used for a wide range of applications, including the three shown in the following table.

Write in the table, a suitable input device, output device, or both needed for each application.

Do not give a monitor, keyboard or mouse in your answers.

Application	Input device	Output device
Capture the text from a paper document, in order that the text can be word processed		
Producing a replica of a small plastic component from a washing machine		
A museum has interactive information facilities throughout the building		

[3]

MARK SCHEME

6(a)	1 mark for each correct row	3												
<table> <tr> <th>Application</th><th>Input device</th><th>Output device</th></tr> <tr> <td>Capture the text from a paper document, in order that the text can be word-processed</td><td>Flatbed scanner / <u>Digital</u> camera</td><td></td></tr> <tr> <td>Producing a replica of a small plastic component from a washing machine</td><td></td><td><u>3D</u> Printer</td></tr> <tr> <td>A museum has interactive information facilities throughout the building</td><td>Touch screen / touch pad / microphone etc.</td><td>Touch screen / speakers etc.</td></tr> </table>			Application	Input device	Output device	Capture the text from a paper document, in order that the text can be word-processed	Flatbed scanner / <u>Digital</u> camera		Producing a replica of a small plastic component from a washing machine		<u>3D</u> Printer	A museum has interactive information facilities throughout the building	Touch screen / touch pad / microphone etc.	Touch screen / speakers etc.
Application	Input device	Output device												
Capture the text from a paper document, in order that the text can be word-processed	Flatbed scanner / <u>Digital</u> camera													
Producing a replica of a small plastic component from a washing machine		<u>3D</u> Printer												
A museum has interactive information facilities throughout the building	Touch screen / touch pad / microphone etc.	Touch screen / speakers etc.												

9608_s17_qp_13_sc_2.a.ii

QUESTION

- (ii) A computer user has a laser printer to print letters and documents. The user also prints digital photographs taken using a digital camera.

State the most suitable type of printer for printing the photographs.

.....[1]

MARK SCHEME

2(a)(ii)	Inkjet printer	1
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9608_s17_qp_13_sc_2.b

QUESTION

- (b) The user is considering the purchase of a new laptop computer. She has read many product reviews and knows that there are different types of internal secondary storage available.

List **two** options for internal secondary storage.

Option 1

Option 2

Describe **one** advantage of one of the options.

Advantage of choosing option 1 / 2 (*circle*)

.....

.....[3]

MARK SCHEME

2(b)	Hard disk drive // HDD	1	3
	Solid state drive //SSD // flash memory	1	
	One from:		
	<i>Hard disk</i>		
	Inexpensive per unit of storage	1	
	<u>Larger</u> storage capacity than flash drive	1	
	<i>Solid state storage</i>		
	No moving parts / noise	1	
	Robust	1	
	Low latency // Fast read/write time	1	

QUESTION

- (c) A student has an old working laptop computer. It has a small capacity internal disk drive with almost all the storage space taken up by the operating system and application programs.

She needs to buy an external storage device to store her data files.

- (i) List **two** suitable devices.

Device 1

Device 2

[2]

MARK SCHEME

2(c)(i)	Two from: External hard drive // External HDD External flash drive // External SSD Pen drive	1 1 1	Max 2
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QUESTION

- (ii) Describe **one** advantage of choosing one of the devices.

Advantage of choosing device 1 / 2 (*circle*)

.....

..... [1]

MARK SCHEME

2(c)(ii)	One from: (<i>External</i>) Hard drive Inexpensive per unit of storage Larger storage capacity than flash drive Or: Pen drive // (<i>External</i>) flash drive No moving parts / noise Low latency // fast access times Robust	1 1 1 1 1 1	Max 1
----------	--	----------------------------	--------------

9608_s17_qp_11_sc_2.a.ii

QUESTION

- (ii) A computer user has a laser printer to print letters and documents. The user also prints digital photographs taken using a digital camera.

State the most suitable type of printer for printing the photographs.

.....[1]

MARK SCHEME

2(a)(ii)	Inkjet printer	1
----------	----------------	---

9608_s17_qp_11_sc_2.b

QUESTION

- (b) The user is considering the purchase of a new laptop computer. She has read many product reviews and knows that there are different types of internal secondary storage available.

List **two** options for internal secondary storage.

Option 1

Option 2

Describe **one** advantage of one of the options.

Advantage of choosing option 1 / 2 (*circle*)

.....

.....[3]

MARK SCHEME

2(b)	Hard disk drive // HDD	1	3
	Solid state drive //SSD // flash memory	1	
	One from:		
	<i>Hard disk</i>		
	Inexpensive per unit of storage	1	
	<u>Larger</u> storage capacity than flash drive	1	
	<i>Solid state storage</i>		
	No moving parts / noise	1	
	Robust	1	
	Low latency // Fast read/write time	1	

9608_w15_qp_13_sc_5.ii

QUESTION

- (ii) When the system detects high activity, operators may need to respond rapidly. A printer is useful for hard copies, but may not be the best way to inform operators.

Give a reason why.

.....
.....[1]

MARK SCHEME

- (ii) – output / hard copy not picked up in good time [1]

9608_w15_qp_13_sc_5.iii

QUESTION

- (iii) Name an alternative output device for this monitoring system and give a reason for your choice.

Output device

Reason for choice

.....
.....[2]

MARK SCHEME

- (iii) 1 mark for name + 1 mark for reason for choice

- speaker / buzzer / alarm bell
- gives control room operators an audible warning
- flashing lights
- gives control room operators an visual warning
- monitor
- use of red / flashing colours on monitor gets the attention of operators [2]

QUESTION

- (iii) Name an alternative output device for this monitoring system and give a reason for your choice.

Output device

Reason for choice

.....

.....

[2]

MARK SCHEME

- (iii) 1 mark for name + 1 mark for reason for choice

- speaker / buzzer / alarm bell
- gives control room operators an audible warning
- flashing lights
- gives control room operators an visual warning
- monitor
- use of red / flashing colours on monitor gets the attention of operators

[2]

QUESTION

- (iii) detecting an intruder in a building

.....[1]

MARK SCHEME

- (iii) sound/acoustic, infrared, pressure, motion, microwave

[1]

QUESTION

- 6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

MARK SCHEME

- 6 (a)

Description of use	Input or output device
input of credit card number into an online form	Keyboard/keypad/numberpad
selection of an option at an airport information kiosk	touch screen
output of a single high quality photograph	ink jet printer
output of several hundred high quality leaflets	laser printer
input of a hard copy image into a computer	scanner

[5]

QUESTION

- 6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

MARK SCHEME

- 6 (a)

Description of use	Input or output device
input of credit card number into an online form	Keyboard/keypad/numberpad
selection of an option at an airport information kiosk	touch screen
output of a single high quality photograph	ink jet printer
output of several hundred high quality leaflets	laser printer
input of a hard copy image into a computer	scanner

[5]

Show understanding of embedded systems

Describe the principal operations of hardware devices

9608_w21_qp_11_sc_4.a

QUESTION

4

Oliver needs to increase the secondary storage on his computer. He already has several hard disks.

(a) Identify **two other** secondary storage devices that Oliver could use. Each device must be different.

1

2 [2]

MARK SCHEME

4(a)	One mark per bullet point to max 2 E.g. • Optical Disc Drive // CD/DVD Drive/Writer • Solid State Drive // USB Flash drive	2
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9608_w21_qp_11_sc_4.b

QUESTION

(b) Oliver needs a scanner to make digital copies of some paper documents.

Describe the basic internal operation of a scanner.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

4(b)	One mark per bullet point to max 4 • Main component of a scanner is a CCD (Charge Couple Device) array • CCD is a collection of light sensitive diodes • Laser beam / light shines onto the source document/barcode • The scanned image reaches the CCD through mirrors and lenses • Sensors detect levels of reflected light • Brighter light results in greater electrical charge • Light intensity is converted (by software) to a digital value	4
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9608_s20_qp_13_sc_2.a

QUESTION

2 Billy has a laser printer.

(a) Complete the following description of the basic internal operation of a laser printer.

The printer uses a and a rotating
to draw the contents of the page on the photosensitive drum as
charge. The is attracted to this charge.

[4]

MARK SCHEME

2(a)	1 mark for each appropriate term The printer uses a laser (beam) and a rotating mirror to draw the contents of the page on the photosensitive drum as an electrostatic charge. The toner is attracted to this charge.	4
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QUESTION

1 Samira is creating an interactive, multimedia presentation for the entrance to her hotel.

(a) The presentation will be on a device that has a resistive touchscreen for user input.

Complete the following paragraph about the basic operation of a resistive touchscreen.

The resistive touchscreen has two layers with between the layers. When a finger touches the screen, the moves to touch the ; this creates a point of contact.

The and position of this point is calculated.

[4]

MARK SCHEME

1(a)	1 mark for each correct term The resistive touchscreen has two layers with a gap / an air gap between the layers. When a finger touches the screen, the top layer moves to touch the bottom layer ; this creates a point of contact. The horizontal and vertical position of this point is calculated.	4
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QUESTION

(b) Kal has built a 3D printer to show students how it works.

(i) The steps 1 to 9 describe the basic internal operation of a 3D printer.

The following five statements are used to complete the sequence of steps.

A	A stepper motor moves the nozzle into position
B	A fan cools the layer
C	The software splits the object into slices
D	The nozzle extrudes the molten plastic
E	The data about the slices is sent to the printer

Write one of the letters **A, B, C, D** or **E** in the appropriate step to complete the sequence.

1. The object is designed using Computer Aided Design (CAD) software
2.
3.
4. The solid plastic is melted and transferred to the nozzle
5.
6.
7. The steps 5 to 6 are repeated until the layer is complete
8.
9. The steps 4 to 8 are repeated for each subsequent layer

[4]

MARK SCHEME

2(b)(i)	1 mark for 1 correct entry 2 marks for 2 correct entries 3 marks for 3 correct entries 4 marks for 5 correct entries - The object is designed using Computer Aided Design (CAD) software C (The software splits the object into slices) E (The data about the slices is sent to the printer) - The solid plastic is melted and transferred to the nozzle A (A stepper motor moves the nozzle into position) D (The nozzle extrudes the molten plastic) - The steps 5 to 6 repeat until the layer is complete B (A fan cools the layer) - The steps 4 to 8 are repeated for each subsequent layer	4
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9608_w19_qp_13_sc_1.a.iii

QUESTION

(iii) The touchscreen uses capacitive technology.

The sequence of steps 1 to 6 describes the internal operation of the touchscreen.

The statements **A**, **B**, **C** and **D** are used to complete the sequence.

A	Charge is drawn to the point of contact.
B	The screen has a layer that stores an electrical charge.
C	There is a change in the electrostatic field.
D	The coordinates are sent to the touchscreen driver.

Write **one** of the letters **A** to **D** in each appropriate row to complete the sequence.

-
- When the user touches the screen
-
-
- The coordinates of the point of contact can be calculated.
-

[2]

MARK SCHEME

1(a)(iii)	1 mark for at least two statements in the correct position, 2 marks for all four statements in correct position. 1 B (The screen has a layer that stores an electrical charge) 2 <i>When the user touches the screen</i> 3 A (Charge is drawn to the point of contact) 4 C (There is a change in the electrostatic field) 5 <i>The coordinates of the point of contact can be calculated</i> 6 D (These coordinates are sent to the touchscreen driver)	2
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9608_w19_qp_12_sc_6.a.i

QUESTION

- 6 Dominic uses a tablet computer to complete work. He records videos of his work to send to his colleagues to watch at a later date.

(a) The tablet computer has input and output devices.

(i) The table lists four devices built into the tablet.

Tick (✓) one or more boxes for each device to identify whether it is an input device, an output device or both.

Device	Input	Output
Touchscreen		
Webcam		
Microphone		
Fingerprint scanner		

[2]

MARK SCHEME

6(a)(i)	1 mark for touchscreen being both 1 mark for remaining 3 devices <table> <tr> <th>Device</th><th>Input</th><th>Output</th></tr> <tr> <td>Touchscreen</td><td>✓</td><td>✓</td></tr> <tr> <td>Webcam</td><td>✓</td><td></td></tr> <tr> <td>Microphone</td><td>✓</td><td></td></tr> <tr> <td>Fingerprint scanner</td><td>✓</td><td></td></tr> </table>	Device	Input	Output	Touchscreen	✓	✓	Webcam	✓		Microphone	✓		Fingerprint scanner	✓		2
Device	Input	Output															
Touchscreen	✓	✓															
Webcam	✓																
Microphone	✓																
Fingerprint scanner	✓																

QUESTION

- (ii) An external speaker is plugged into the tablet computer.

The sequence of steps 1 to 7 describes the internal operation of the speaker.

The statements **A**, **B**, **C**, **D** and **E** are used to complete the sequence.

Letter	Statement
A	Changes in the audio signal cause the direction of the electrical current to change. This determines the polarity of the electromagnet.
B	The vibration creates sound waves.
C	An electric current is sent to the speaker.
D	The electromagnet is repelled by, or attracted to the permanent magnet.
E	The electric current passes through the coil.

Write **one** of the letters **A** to **E** in each appropriate row to complete the sequence.

- 1
- 2
- 3 The current in the coil creates an electromagnetic field.
- 4
- 5
- 6 The movement of the coil causes the diaphragm to vibrate.
- 7

[4]

MARK SCHEME

6(a)(ii)	1 mark for any 1 correct letter in the correct position 2 marks for any 2 correct letters in the correct positions 3 marks for any 3 correct letters in the correct positions 4 marks for 5 correct letters in the correct positions C An electric current is sent to the speaker. E The electric current passes through the coil. <i>The current in the coil creates an electromagnetic field.</i> A Changes in the audio signal cause the direction of the electrical current to change. This determines the polarity of the electromagnet. D The electromagnet is repelled by, or attracted to the permanent magnet. <i>The movement of the coil causes the diaphragm to vibrate.</i> B The vibration creates sound waves.	4
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QUESTION

(ii) Describe solid state memory.

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.....

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.....

..... [3]

MARK SCHEME

6(b)(ii)	1 mark per bullet point to max 3 No moving parts Solid state memory is non-volatile Makes use of blocks / arrays of ... Semiconductors // NAND gates // NOR gates // transistors // integrated circuits SSD Controller manages the components Uses a grid of columns and rows that has two transistors at each intersection One transistor is called a floating gate The second transistor is called the control gate Memory cells store voltages which can represent either a 0 or a 1 Essentially the movement of electrons is controlled to read/write Not possible to overwrite existing data // it is necessary to first erase the old data then write the new data in the same location	3
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QUESTION

(iv) Describe the internal operation of a trackerball mouse.

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.....

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.....

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..... [3]

MARK SCHEME

2(a)(iv)	1 mark per bullet point to max 3 The ball touches horizontal and vertical rollers When the ball rotates / moves ... one or both of the rollers rotate as well Each roller connects to a shaft which spins a disk with holes Infrared beams shine through the holes in the disks As the ball moves the roller the beam is broken by the space between the holes ... creating pulses of light The distance and/or speed of the mouse is determined from the rate of the pulses ... by an on-board processor chip // by driver software in the computer	3
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9608_s19_qp_12_sc_4.a.i

QUESTION

- 4 Shazia is creating a computer program that will be released to the public. The program includes a video.

(a) Shazia uses a microphone to record a sound track for the video.

(i) Describe the internal operation of a microphone.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

4(a)(i)	1 mark per bullet point to max 3 - The microphone has a diaphragm / ribbon (accept equivalent) - The incoming sound <u>waves</u> cause vibrations (of the diaphragm) ...causing a coil to move past a magnet (dynamic microphone) // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) etc. - An electrical signal is produced	3
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QUESTION

- (ii) The script for the sound track is printed using a laser printer.

Describe the internal operation of a laser printer.

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.....

.....

.....

.....

..... [3]

MARK SCHEME

4(a)(ii)	1 mark per bullet to max 3 • The revolving drum is initially given an electrical charge • A laser beam (bounces off moving mirrors) <u>scans back and forth</u> across the drum • ...discharging certain points (i.e. 'drawing' the letters and images to be printed as a pattern of electrical charges) • The drum is coated with oppositely charged toner (which only sticks to charged areas) • The drum rolls over electro-statically <u>charged</u> paper // Electro-statically <u>charged</u> paper is fed (towards the drum) • The 'pattern' on the drum is transferred to the paper • The paper is passed through the fuser to seal the image • The electrical charge is removed from the drum // the excess toner is collected	3
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QUESTION

1 A student is creating a short video and needs to record music to play in the background.

(a) The student uses a microphone to capture the music.

Explain how the microphone captures the music.

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.....

.....

.....[3]

MARK SCHEME

1(a)	1 mark per bullet point to max 3 The microphone has a diaphragm The incoming sound waves cause vibrations ... causing a coil to move past a magnet (dynamic microphone) // changing the capacitance (condenser microphone) An electric current is generated / changed	3
------	---	----------

QUESTION

(c) Give the most appropriate secondary storage device for this computer system.

Describe **two** reasons for your choice.

Device

Reason 1

.....

.....

.....

Reason 2

.....

.....

.....

[5]

MARK SCHEME

7(c)	1 mark for device (Internal) hard <u>drive</u> / solid state <u>drive</u> 1 mark per bullet. Max 2 marks for each reason, max two reasons e.g. Hard drive - Large capacity... to store videos / images / sound files with large file sizes - Reasonably fast access speed... Users will not have to wait for videos to load - Inexpensive per unit storage... If a large number of needed for different exhibits, the cost can be kept low - Does not need to be moved ... So moving parts unlikely to be damaged - Slower degradation of data ... So will last longer / be more reliable under heavy use Solid state - Large capacity... To store videos/images/sound files with large file sizes - Fast access speed... Users will not have to wait for videos to load - Reliable... Can be dropped/damaged and will likely still work / no moving parts - Quiet... No moving parts	5
------	--	----------

QUESTION

Instruction		Op code (binary)	Explanation
Op code (mnemonic)	Operand		
LDM	#n	1100 0001	Immediate addressing. Load number <i>n</i> to ACC.
LDD	<address>	1100 0010	Direct addressing. Load the contents of the given address to ACC.
LDV	#n	1100 0011	Relative addressing. Move to the address <i>n</i> locations from the address of the current instruction. Load the contents of this address to ACC.
STO	<address>	1100 0100	Store the contents of ACC at the given address.
DEC		1100 0101	Decrement the contents of ACC.
OUTCH		1100 0111	Output the character corresponding to the ASCII character code in ACC.
JNE	<address>	1110 0110	Following a compare instruction, jump to <address> if the compare was False.
JMP	<address>	1110 0011	(Unconditionally) jump to the given address.
CMP	#n	1110 0100	Compare the contents of ACC with number <i>n</i> .

5 A Personal Computer (PC) has a number of input and output devices.

(a) (i) Name **three** components of a speaker.

1

2

3

[3]

MARK SCHEME

5(a)(i)	Three marks from: Diaphragm / cone (Voice) coil of wire Spider / Suspension (Permanent) Magnet Basket Dust cap Outer frame	3
---------	---	----------

QUESTION

(ii) Explain the basic internal operation of a speaker.

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

5(a)(ii)	Four marks from: Takes an electrical signal and translates it into physical vibrations to create sound waves An electric current in the coil creates an electro-magnetic field Changes in the audio signal causes the direction of the electric current to change The direction of the current determines the polarity of the electro-magnet // changing the direction of the current changes the direction of the polarity of the electro-magnet The electro-magnet is repelled by or attracted to the permanent magnet Causing the coil to vibrate The movement of the coil causes the cone / diaphragm to vibrate That vibration is transmitted to the air in front of the cone / diaphragm as sound waves The amount of movement will determine the frequency and amplitude of the sound wave produced	Max 4
----------	---	--------------

QUESTION

- 6 (a) A personal computer (PC) is extensively used for a wide range of applications, including the three shown in the following table.

Write in the table, a suitable input device, output device, or both needed for each application.

Do not give a monitor, keyboard or mouse in your answers.

Application	Input device	Output device
Capture the text from a paper document, in order that the text can be word processed		
Producing a replica of a small plastic component from a washing machine		
A museum has interactive information facilities throughout the building		

[3]

MARK SCHEME

6(a)	1 mark for each correct row	3												
	<table> <tr> <th>Application</th><th>Input device</th><th>Output device</th></tr> <tr> <td>Capture the text from a paper document, in order that the text can be word-processed</td><td>Flatbed scanner / <u>Digital</u> camera</td><td></td></tr> <tr> <td>Producing a replica of a small plastic component from a washing machine</td><td></td><td><u>3D</u> Printer</td></tr> <tr> <td>A museum has interactive information facilities throughout the building</td><td>Touch screen / touch pad / microphone etc.</td><td>Touch screen / speakers etc.</td></tr> </table>	Application	Input device	Output device	Capture the text from a paper document, in order that the text can be word-processed	Flatbed scanner / <u>Digital</u> camera		Producing a replica of a small plastic component from a washing machine		<u>3D</u> Printer	A museum has interactive information facilities throughout the building	Touch screen / touch pad / microphone etc.	Touch screen / speakers etc.	
Application	Input device	Output device												
Capture the text from a paper document, in order that the text can be word-processed	Flatbed scanner / <u>Digital</u> camera													
Producing a replica of a small plastic component from a washing machine		<u>3D</u> Printer												
A museum has interactive information facilities throughout the building	Touch screen / touch pad / microphone etc.	Touch screen / speakers etc.												

.....[4]

MARK SCHEME

6(b)	1 Mark per bullet to max 4	4
	The hard disk has one or more platters made of aluminium or glass Each surface of the platter/disk is ferrous-oxide which is capable of being magnetised The platters/disks are mounted on a central spindle The disks are rotated at high-speed Each surface of the disk has a read/write head mounted on an arm positioned just above the surface Electronic circuits control the movement of the arm and hence the heads The surface of the platter/disk is divided into <u>concentric</u> tracks and sectors One track in one sector is the basic unit of storage called a block The data is encoded as a magnetic pattern for each block When writing to disk, a variation in the current in the head produces a variation in magnetic field on the disk When reading from disk, a variation in magnetic field produces a variation in current through the head	

QUESTION

- 2 (a) (i) The following sequence of steps (1 to 7) describe how a single page is printed on a laser printer.

The statements A, B, C and D are used to complete the sequence.

A	The paper passes through a fuser, which heats up the paper. The toner melts and forms a permanent image on the paper.
B	The electrical charge is removed from the drum and the excess toner is collected.
C	The image is converted on the drum into an electrostatic charge.
D	The oppositely-charged paper picks up the toner particles from the drum. After picking up the toner, the paper is discharged to stop it clinging to the drum.

Complete the sequence by writing one of the letters **A**, **B**, **C** or **D** on the appropriate row.

1. A laser beam and a rotating mirror are used to draw an image of the page on the photosensitive drum.
2.
3. Electrostatic charge attracts toner.
4. The charged paper is rolled against the drum.
5.
6.
7.

[3]

MARK SCHEME

2(a)(i)	1	A laser beam and a rotating mirror are used to draw an image of the page on the photosensitive drum.	3
	2	C // The image is converted on the drum into an electrostatic charge.	
	3	Electrostatic charge attracts toner.	
	4	The charged paper is rolled against the drum.	
	5	D // The oppositely-charged paper picks up the toner particles from the drum. After picking up the toner, the paper is discharged to stop it clinging to the drum.	
	6	A // The paper passes through a fuser, which heats up the paper. The toner melts and forms a permanent image on the paper.	
	7	B // The electrical charge is removed from the drum and the excess toner is collected.	
C in the correct place			1
DA,			1
AB			1

9608_s17_qp_13_sc_2.a.ii

QUESTION

- (ii) A computer user has a laser printer to print letters and documents. The user also prints digital photographs taken using a digital camera.

State the most suitable type of printer for printing the photographs.

.....[1]

MARK SCHEME

2(a)(ii)	Inkjet printer	1
----------	----------------	---

9608_s17_qp_13_sc_2.b

QUESTION

- (b) The user is considering the purchase of a new laptop computer. She has read many product reviews and knows that there are different types of internal secondary storage available.

List **two** options for internal secondary storage.

Option 1

Option 2

Describe **one** advantage of one of the options.

Advantage of choosing option 1 / 2 (*circle*)

.....

.....[3]

MARK SCHEME

2(b)	Hard disk drive // HDD	1	3
	Solid state drive //SSD // flash memory	1	
	One from:		
	<i>Hard disk</i>		
	Inexpensive per unit of storage	1	
	<u>Larger</u> storage capacity than flash drive	1	
	<i>Solid state storage</i>		
	No moving parts / noise	1	
	Robust	1	
	Low latency // Fast read/write time	1	

QUESTION

- 2 (a) The first column of the following table gives features of different types of printer.

Put a tick (✓) in the cells to show which features describe a laser and an inkjet printer.

	Type of printer	
	Laser	Inkjet
Impact printer		
Non-impact printer		
Line printer		
Page printer		

[2]

MARK SCHEME

2(a)	One mark for each pair of rows	2																	
	<table> <tr> <th rowspan="2"></th><th colspan="2">Type of printer</th></tr> <tr> <th>Laser</th><th>Inkjet</th></tr> <tr> <td>Impact printer</td><td></td><td></td></tr> <tr> <td>Non-impact printer</td><td>✓</td><td>✓</td></tr> <tr> <td>Line printer</td><td></td><td>✓</td></tr> <tr> <td>Page printer</td><td>✓</td><td></td></tr> </table>		Type of printer		Laser	Inkjet	Impact printer			Non-impact printer	✓	✓	Line printer		✓	Page printer	✓		1 1
	Type of printer																		
	Laser	Inkjet																	
Impact printer																			
Non-impact printer	✓	✓																	
Line printer		✓																	
Page printer	✓																		

Describe how each component is used when printing a page.

(i) Print head [5]

MARK SCHEME

2(b)(i)	Five from:		Max 5
	The print head contains a large number of very small nozzles	1	
	Ink is fed to each nozzle from a reservoir	1	
	The print head fires <u>droplets</u> of ink onto the paper	1	
	The print head moves horizontally across the paper	1	
	Either:		
	Tiny resistors create heat inside each nozzle	1	
	The heat vaporises ink to create a bubble	1	
	When the bubble pops the ink is deposited on the page	1	
	The collapsing bubble creates a partial vacuum in the nozzle	1	
	And ink is drawn from the reservoir ready for printing the next dot	1	
	Or:		
	There is a piezo crystal at the back of the ink reservoir of each nozzle	1	
	The crystal vibrates when it receives a tiny electric charge	1	
	Ink is forced out of the nozzle by the inward vibration	1	
	The outward vibration creates a partial vacuum in the nozzle	1	
	Replacement ink is pulled into the reservoir	1	

9608_s17_qp_12_sc_2.b.ii

QUESTION

(ii) Stepper motor

.....

[2]

MARK SCHEME

2(b)(ii)	Two from: The (print head) stepper motor is connected to the print head by a belt 1 The (print head) stepper motor moves the print head across the paper 1 The (parking) stepper motor parks the print head assembly when not in use 1 The (paper feed)stepper motor turns the rollers that provide the paper feed // The (paper feed)stepper motor moves the paper in small increments 1	Max 2
----------	---	--------------

9608_s17_qp_12_sc_2.c.i

QUESTION

(c) A student has an old working laptop computer. It has a small capacity internal disk drive with almost all the storage space taken up by the operating system and application programs.

She needs to buy an external storage device to store her data files.

(i) List **two** suitable devices.

Device 1
 Device 2
 [2]

MARK SCHEME

2(c)(i)	Two from: External hard drive // External HDD 1 External flash drive // External SSD 1 Pen drive 1	Max 2
---------	--	--------------

QUESTION

(ii) Describe **one** advantage of choosing one of the devices.

Advantage of choosing device 1 / 2 (*circle*)

.....
[1]

MARK SCHEME

2(c)(ii)	One from: <i>(External) Hard drive</i> Inexpensive per unit of storage Larger storage capacity than flash drive Or: <i>Pen drive // (External) flash drive</i> No moving parts / noise Low latency // fast access times Robust	 1 1 1 1 1	Max 1
----------	--	---	--------------

QUESTION

- 2 (a) (i) The following sequence of steps (1 to 7) describe how a single page is printed on a laser printer.

The statements A, B, C and D are used to complete the sequence.

A	The paper passes through a fuser, which heats up the paper. The toner melts and forms a permanent image on the paper.
B	The electrical charge is removed from the drum and the excess toner is collected.
C	The image is converted on the drum into an electrostatic charge.
D	The oppositely-charged paper picks up the toner particles from the drum. After picking up the toner, the paper is discharged to stop it clinging to the drum.

Complete the sequence by writing one of the letters **A**, **B**, **C** or **D** on the appropriate row.

1. A laser beam and a rotating mirror are used to draw an image of the page on the photosensitive drum.
2.
3. Electrostatic charge attracts toner.
4. The charged paper is rolled against the drum.
5.
6.
7.

[3]

MARK SCHEME

2(a)(i)	1	A laser beam and a rotating mirror are used to draw an image of the page on the photosensitive drum.	3
	2	C // The image is converted on the drum into an electrostatic charge.	
	3	Electrostatic charge attracts toner.	
	4	The charged paper is rolled against the drum.	
	5	D // The oppositely-charged paper picks up the toner particles from the drum. After picking up the toner, the paper is discharged to stop it clinging to the drum.	
	6	A // The paper passes through a fuser, which heats up the paper. The toner melts and forms a permanent image on the paper.	
	7	B // The electrical charge is removed from the drum and the excess toner is collected.	
C in the correct place			1
DA,			1
AB			1

9608_s17_qp_11_sc_2.a.ii

QUESTION

- (ii) A computer user has a laser printer to print letters and documents. The user also prints digital photographs taken using a digital camera.

State the most suitable type of printer for printing the photographs.

.....[1]

MARK SCHEME

2(a)(ii)	Inkjet printer	1
----------	----------------	---

9608_s17_qp_11_sc_2.b

QUESTION

- (b) The user is considering the purchase of a new laptop computer. She has read many product reviews and knows that there are different types of internal secondary storage available.

List **two** options for internal secondary storage.

Option 1

Option 2

Describe **one** advantage of one of the options.

Advantage of choosing option 1 / 2 (*circle*)

.....

.....[3]

MARK SCHEME

2(b)	Hard disk drive // HDD	1	3
	Solid state drive //SSD // flash memory	1	
	One from:		
	<i>Hard disk</i>		
	Inexpensive per unit of storage	1	
	<u>Larger</u> storage capacity than flash drive	1	
	<i>Solid state storage</i>		
	No moving parts / noise	1	
	Robust	1	
	Low latency // Fast read/write time	1	

QUESTION

4 Describe the basic internal operation of the following devices:

(i) Keyboard

.....

.....

.....

.....[2]

MARK SCHEME

4 (i) **Keyboard**

Any **two** from:

- Uses switches and circuits to translate keystrokes into signals the computer can understand
- The key matrix is a grid of circuits / three layers of plastic underneath the keys
- Each circuit is broken beneath the key / middle layer contains holes
- When key pressed, a circuit is made / completed and a signal is sent
- Processor compares location of signal from key matrix to a character map stored on ROM
- A character code for each key press is saved in a keyboard buffer

[2]

QUESTION

(ii) Optical disc

.....

.....

.....

.....[2]

MARK SCHEME

(ii) Optical Disc

Any **two** from:

- Drive motor is used to spin the disc
 - Tracking mechanism moves the laser assembly
 - A lens focuses the laser onto the disc
 - Laser beam is shone onto disc to read / write
 - Surface of disc has a reflective metal layer / phase change metal alloy
 - Track(s) on the disc have sequence of pits and lands / amorphous and crystalline state
 - Reflected light is then encoded as a bit pattern
- [2]

9608_w16_qp_13_sc_4.iii

QUESTION

(iii) Optical mouse

.....
.....
.....
.....[2]

MARK SCHEME

(iii) Optical mouse

Any **two** from:

- Laser / light shines onto a surface
 - Through a (polished) ring at the base
 - The light is reflected from the surface through the ring
 - Sensor detects reflected light
 - Capturing details / photograph of surface (under the ring)
 - At about 1500 times per second
 - As the mouse moves the sensor detects changes in the surface detail / photograph
 - Which are translated into movement (change of x and y co-ordinates)
 - The processor/software updates the position of the cursor on the screen
- [2]

QUESTION

(iv) Scanner

.....

.....

.....

.....[2]

MARK SCHEME

(iv) **Scanner**

Any **two** from:

- Main component of a scanner is a CCD array
- CCD is a collection of light sensitive diodes
- Laser beam / light is shone onto the source document/barcode
- The scanned image reaches the CCD through mirrors and lenses
- Sensors detect levels of reflected light
- Brighter light results in greater electrical charge
- Light intensity is converted (by software) to a digital value

[2]

QUESTION

- 3 When an application program requests a file stored on a hard disk, the computer system reads the file. Use the statement labels A to H to complete the sequence of steps that describe how this happens.

Label	Statement
A	When the hard disk drive has read the file, it generates an interrupt.
B	While the file continues, the head reads successive clusters of sectors from the disk and writes data into the disk buffer.
C	The head reads the first cluster of sectors from disk and writes data into the disk buffer.
D	The head moves to the correct track.
E	The operating system transfers the contents of the disk buffer to the application program's data memory.
F	In the relevant directory file, the operating system looks up the track and sector where the file begins.
G	Application program passes file read request to the operating system.
H	The hard disk drive waits for the correct sector to arrive under the head.

1. The application program executes a statement to read a file.
2.
3. The operating system begins to spin the hard disk, if it is not currently spinning.
4.
5.
6.
7.
8.
9.
10.

[8]

MARK SCHEME

3 **ONE** mark for each letter in the correct place.

Then **ONE** mark for any pair of letters in the correct order, but not in the correct place

- | | |
|----|---|
| 1 | The application program executes a statement to read a file. |
| 2 | G |
| 3 | The operating system begins to spin the hard disk, if it is not currently spinning. |
| 4 | F |
| 5 | D |
| 6 | H |
| 7 | C |
| 8 | B |
| 9 | A |
| 10 | E |

[8]

9608_w16_qp_11_sc_4.i

QUESTION

4 Describe the basic internal operation of the following devices:

(i) Keyboard

.....
.....
.....
.....[2]

MARK SCHEME

4 (i) Keyboard

Any **two** from:

- Uses switches and circuits to translate keystrokes into signals the computer can understand
- The key matrix is a grid of circuits / three layers of plastic underneath the keys
- Each circuit is broken beneath the key / middle layer contains holes
- When key pressed, a circuit is made / completed and a signal is sent
- Processor compares location of signal from key matrix to a character map stored on ROM
- A character code for each key press is saved in a keyboard buffer

[2]

9608_w16_qp_11_sc_4.ii

QUESTION

(ii) Optical disc

.....

.....

.....

.....[2]

MARK SCHEME

(ii) Optical Disc

Any **two** from:

- Drive motor is used to spin the disc
- Tracking mechanism moves the laser assembly
- A lens focuses the laser onto the disc
- Laser beam is shone onto disc to read / write
- Surface of disc has a reflective metal layer / phase change metal alloy
- Track(s) on the disc have sequence of pits and lands / amorphous and crystalline state
- Reflected light is then encoded as a bit pattern

[2]

QUESTION

(iii) Optical mouse

.....

.....

.....

.....[2]

MARK SCHEME

(iii) **Optical mouse**

Any **two** from:

- Laser / light shines onto a surface
- Through a (polished) ring at the base
- The light is reflected from the surface through the ring
- Sensor detects reflected light
- Capturing details / photograph of surface (under the ring)
- At about 1500 times per second
- As the mouse moves the sensor detects changes in the surface detail / photograph
- Which are translated into movement (change of x and y co-ordinates)
- The processor/software updates the position of the cursor on the screen

[2]

QUESTION

(iv) Scanner

.....

.....

.....

.....[2]

MARK SCHEME

(iv) Scanner

Any **two** from:

- Main component of a scanner is a CCD array
- CCD is a collection of light sensitive diodes
- Laser beam / light is shone onto the source document/barcode
- The scanned image reaches the CCD through mirrors and lenses
- Sensors detect levels of reflected light
- Brighter light results in greater electrical charge
- Light intensity is converted (by software) to a digital value

[2]

9608_s15_qp_13_sc_4.c

QUESTION

(c) DVD-RAM and flash memory are two examples of storage devices.

Describe **two** differences in how they operate.

.....

.....

.....

.....

..... [2]

MARK SCHEME

(c) **one** mark for DVD-RAM, **one** mark for flash memory.

DVD-RAM

- data is stored/written using lasers/optical media
- DVD-RAM uses phase changing recording, in which varying laser intensities cause targeted areas in the phase change recording layer to alternate between an amorphous and a crystalline state.
- uses a rotating disk with concentric tracks
- allows read and write operation to occur simultaneously

flash memory

- most are NAND-based flash memory
- there are no moving parts
- uses a grid of columns and rows that has two transistors at each intersection
- one transistor is called a floating gate
- the second transistor is called the control gate
- memory cells store voltages which can represent either a 0 or a 1
- essentially the movement of electrons is controlled to read/write
- not possible to over-write existing data; it is necessary to first erase the old data then write the new data in the same location

[2]

9608_s15_qp_12_sc_6.a

QUESTION

6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

MARK SCHEME

6 (a)	<table> <tr> <th data-bbox="239 235 917 336">Description of use</th><th data-bbox="917 235 1461 336">Input or output device</th></tr> <tr> <td data-bbox="239 336 917 470">input of credit card number into an online form</td><td data-bbox="917 336 1461 470">Keyboard/keypad/numberpad</td></tr> <tr> <td data-bbox="239 470 917 638">selection of an option at an airport information kiosk</td><td data-bbox="917 470 1461 638">touch screen</td></tr> <tr> <td data-bbox="239 638 917 772">output of a single high quality photograph</td><td data-bbox="917 638 1461 772">ink jet printer</td></tr> <tr> <td data-bbox="239 772 917 907">output of several hundred high quality leaflets</td><td data-bbox="917 772 1461 907">laser printer</td></tr> <tr> <td data-bbox="239 907 917 1030">input of a hard copy image into a computer</td><td data-bbox="917 907 1461 1030">scanner</td></tr> </table>	Description of use	Input or output device	input of credit card number into an online form	Keyboard/keypad/numberpad	selection of an option at an airport information kiosk	touch screen	output of a single high quality photograph	ink jet printer	output of several hundred high quality leaflets	laser printer	input of a hard copy image into a computer	scanner
Description of use	Input or output device												
input of credit card number into an online form	Keyboard/keypad/numberpad												
selection of an option at an airport information kiosk	touch screen												
output of a single high quality photograph	ink jet printer												
output of several hundred high quality leaflets	laser printer												
input of a hard copy image into a computer	scanner												
[5]													

9608_s15_qp_11_sc_6.a

QUESTION

6 (a) Name the **most** suitable input or output device for each of the following uses.

Give a different device in **each** case.

Description of use	Input or output device
input of credit card number into an online form	
selection of an option at an airport information kiosk	
output of a single high-quality photograph	
output of several hundred high-quality leaflets	
input of a hard copy image into a computer	

[5]

MARK SCHEME

6 (a)

Description of use	Input or output device
input of credit card number into an online form	Keyboard/keypad/numberpad
selection of an option at an airport information kiosk	touch screen
output of a single high quality photograph	ink jet printer
output of several hundred high quality leaflets	laser printer
input of a hard copy image into a computer	scanner

[5]

Show understanding of the use of buffers

9608_w16_qp_13_sc_6

QUESTION

- 6** A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 6** Any **four** from:

- User needs high-speed broadband (connection)
- Data is streamed to a buffer (in the computer)
- Buffering stops video pausing as bits streamed
- As buffer is emptied, it fills up again so that viewing is continuous
- Actual playback is (a few seconds) behind the time the data is received by computer [4]

QUESTION

- 3 When an application program requests a file stored on a hard disk, the computer system reads the file. Use the statement labels A to H to complete the sequence of steps that describe how this happens.

Label	Statement
A	When the hard disk drive has read the file, it generates an interrupt.
B	While the file continues, the head reads successive clusters of sectors from the disk and writes data into the disk buffer.
C	The head reads the first cluster of sectors from disk and writes data into the disk buffer.
D	The head moves to the correct track.
E	The operating system transfers the contents of the disk buffer to the application program's data memory.
F	In the relevant directory file, the operating system looks up the track and sector where the file begins.
G	Application program passes file read request to the operating system.
H	The hard disk drive waits for the correct sector to arrive under the head.

- The application program executes a statement to read a file.
-
- The operating system begins to spin the hard disk, if it is not currently spinning.
-
-
-
-
-
-
-

[8]

MARK SCHEME

3 **ONE** mark for each letter in the correct place.

Then **ONE** mark for any pair of letters in the correct order, but not in the correct place

- | | |
|----|---|
| 1 | The application program executes a statement to read a file. |
| 2 | G |
| 3 | The operating system begins to spin the hard disk, if it is not currently spinning. |
| 4 | F |
| 5 | D |
| 6 | H |
| 7 | C |
| 8 | B |
| 9 | A |
| 10 | E |

[8]

9608_w16_qp_11_sc_6

QUESTION

6 A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

6 Any **four** from:

- User needs high-speed broadband (connection)
 - Data is streamed to a buffer (in the computer)
 - Buffering stops video pausing as bits streamed
 - As buffer is emptied, it fills up again so that viewing is continuous
 - Actual playback is (a few seconds) behind the time the data is received by computer
- [4]

Explain the differences between Random Access Memory (RAM) and Read Only Memory (ROM)

9608_s20_qp_13_sc_2.b

QUESTION

(b)

The laser printer has both RAM and ROM.

Describe the purpose of RAM and ROM in the **laser printer**.

RAM

ROM

[4]

MARK SCHEME

2(b)	1 mark per bullet point. Max 2 for RAM, max 2 for ROM RAM - Stores currently running parts of the printer software - Stores the data being printed // contents of buffer - Stores current progress of printing - Stores the data about the printer, e.g. toner levels ROM - Stores the printer operating software // OS for the printer - Stores the boot-up/start-up instructions for the printer - Printer fonts stored in ROM	4
------	---	---

QUESTION

- (ii) The 3D printer has both RAM and ROM.

Describe the purpose of RAM and ROM in a **3D printer**.

RAM

.....

.....

.....

ROM

.....

.....

.....

[4]

MARK SCHEME

2(b)(ii)	1 mark per bullet point. Max 3 for RAM, max 2 for ROM RAM • Stores currently running parts of the 3D printer software • Stores the data about the layers being printed // contents of buffer • Stores current progress of printing • Stores the data about the printer, e.g. Plastic levels, nozzle position ROM • Stores the operating software for the 3D printer // OS for the 3D printer • Stores the boot-up/start-up instructions for the 3D printer	4
----------	--	----------

QUESTION

- (c) The tablet computer has RAM and ROM memory.

State the purpose of RAM and ROM memory in the computer.

RAM

.....

ROM

.....

[2]

MARK SCHEME

6(c)	1 mark per bullet point to max 2 RAM stores currently running parts of files / programs / processes / OS ROM stores boot up instructions / OS kernel // data permanently // store the firmware for the tablet	2
------	---	----------

9608_s18_qp_13_sc_7.d

QUESTION

(d) This computer system has Random Access Memory (RAM) and Read Only Memory (ROM).
State what will be stored in RAM and ROM for this computer system.

RAM

.....

ROM

.....

[2]

MARK SCHEME

7(d)	1 mark for a correct reason for RAM, 1 mark for a correct reason for ROM RAM Currently running data / video / music / images / software ROM Boot up instructions / OS kernel	2
------	--	----------

9608_s18_qp_11_sc_7.b.i

QUESTION

(b) The games console has random access memory (RAM) and read only memory (ROM).

(i) State **two** differences between RAM and ROM.

Difference 1

.....

Difference 2

.....

[2]

MARK SCHEME

7(b)(i)	1 mark for each difference to max 2 RAM is volatile and ROM is non-volatile RAM can change and ROM (usually) can't be changed ROM is read only, RAM is read/write	2
---------	--	----------

9608_s18_qp_11_sc_7.b.ii

QUESTION

(ii) Give **one** use for RAM in the games console.

.....

.....[1]

MARK SCHEME

7(b)(ii)	1 mark for example e.g. Current game Currently running processes Current graphics/sound	1
----------	--	----------

9608_s18_qp_11_sc_7.b.iii

QUESTION

(iii) Give **one** use for ROM in the games console.

.....

.....[1]

MARK SCHEME

7(b)(iii)	1 mark for an example e.g. Start-up instructions / boot program Kernel of Operating System	1
-----------	--	----------

QUESTION

6 (a) Describe **two** differences between RAM and ROM.

- 1
-
- 2
- [2]

MARK SCHEME

6 (a) **ONE** mark for each difference from the bullet points below.

- RAM loses content when power turned off / volatile memory / temporary memory
ROM does not lose content when power turned off / non-volatile memory / permanent memory
- Data in RAM can be altered / deleted / read from and written to
ROM is read only / cannot be changed / altered / deleted
- RAM stores files / data / operating system currently in use
ROM is used to store BIOS / bootstrap / pre-set instructions

[2]

QUESTION

(b) Describe **three** differences between RAM and ROM.

-
-
-
-
-
- [3]

MARK SCHEME

(b) maximum of **two marks** for RAM and maximum of **two marks** for ROM

RAM

- loses contents when power turned off/volatile memory/temporary memory
- stores files/data/operating system currently in use
- data can be altered/deleted/read from and written to
- memory size is often larger than ROM

ROM

- doesn't lose contents when power turned off/non-volatile memory/permanent memory
- cannot be changed/altered/deleted/read only
- can be used to store BIOS/bootstrap

[3]

Explain the differences between Static RAM (SRAM) and Dynamic RAM (DRAM)

9608_w21_qp_11_sc_4.c

QUESTION

- (c) Oliver wants to upgrade the RAM in his computer. He is not sure whether his computer has Static RAM (SRAM) or Dynamic RAM (DRAM).

Draw **one or more** lines from each type of RAM to its appropriate description(s).

Type of RAM	Description
	Is less expensive to manufacture
SRAM	Needs to be refreshed
	Has more complex circuitry
DRAM	Is often used as cache
	Has faster access time

[2]

MARK SCHEME

4(c)	One mark for correct lines from each type of RAM Type of RAM Description <pre> graph LR SRAM[SRAM] --- L1[Is less expensive to manufacture] SRAM --- L2[Has more complex circuitry] SRAM --- L3[Is often used as cache] DRAM[DRAM] --- L4[Needs to be refreshed] DRAM --- L3 DRAM --- L5[Has faster access time] </pre>	2
------	---	---

9608_w19_qp_13_sc_1.b.ii

QUESTION

- (ii) The self-checkout machines use Static RAM (SRAM) for their cache.

The following table has statements about SRAM or Dynamic RAM (DRAM).

Tick (✓) **one** box in each row to identify whether the statement is about SRAM or DRAM.

Statement	SRAM	DRAM
More expensive to make		
Requires refreshing (recharging)		
Made from flip-flops		

[2]

MARK SCHEME

1(b)(ii)	1 mark for at least one correct row, 2 marks for all three correct rows	2												
<table><tr><th>Statement</th><th>SRAM</th><th>DRAM</th></tr><tr><td>More expensive to make</td><td>✓</td><td></td></tr><tr><td>Requires refreshing (recharging)</td><td></td><td>✓</td></tr><tr><td>Made from flip-flops</td><td>✓</td><td></td></tr></table>			Statement	SRAM	DRAM	More expensive to make	✓		Requires refreshing (recharging)		✓	Made from flip-flops	✓	
Statement	SRAM	DRAM												
More expensive to make	✓													
Requires refreshing (recharging)		✓												
Made from flip-flops	✓													

9608_s19_qp_12_sc_4.c

QUESTION

(c) Shazia’s computer has Dynamic RAM (DRAM) and Static RAM (SRAM).

Explain the differences between Dynamic RAM and Static RAM.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

4(c)	1 mark per bullet to max 4 • DRAM has to be refreshed / charged and SRAM does not require a refresh • DRAM uses a single transistor and capacitor and SRAM uses more than one transistor • DRAM stores each bit as a charge and in SRAM each bit is stored using a flip-flop/latch • DRAM requires higher power consumption under low levels of access, (which is significant when used in battery-powered devices because it requires more circuitry for refreshing) // SRAM uses less power (no need to refresh) • DRAM less expensive to purchase (requires fewer transistors) // SRAM is more expensive to buy (as it requires more transistors) • DRAM has slower <u>access</u> time/speed (because it needs to be refreshed) // SRAM has faster <u>access</u> times • DRAM can have higher storage/bit/data <u>density</u> // SRAM has lower storage/bit/data <u>density</u> • DRAM used in main memory and SRAM used in cache memory	4
------	--	----------

9608_w18_qp_13_sc_6.c

QUESTION

(c) There are two types of RAM: dynamic RAM (DRAM) and static RAM (SRAM).

The following table shows **five** statements about DRAM and SRAM.

Tick (✓) **one** box in each row to indicate whether the statement applies to DRAM or SRAM.

Statement	DRAM	SRAM
Does not need to be refreshed as the circuit holds the data while the power supply is on		
Mainly used in cache memory of processors where speed is important		
Has less complex circuitry		
Requires higher power consumption under low levels of access, which is significant when used in battery-powered devices		
Requires data to be refreshed occasionally so it retains the data		

[5]

MARK SCHEME

6(c)	1 mark per row <table border="1"> <thead> <tr> <th data-bbox="295 286 1109 347">Statement</th><th data-bbox="1109 286 1225 347">DRAM</th><th data-bbox="1225 286 1342 347">SRAM</th></tr> </thead> <tbody> <tr> <td data-bbox="295 347 1109 448">Does not need to be refreshed as the circuit holds the data while the power supply is on</td><td data-bbox="1109 347 1225 448"></td><td data-bbox="1225 347 1342 448">✓</td></tr> <tr> <td data-bbox="295 448 1109 548">Mainly used in cache memory of processors where speed is important</td><td data-bbox="1109 448 1225 548"></td><td data-bbox="1225 448 1342 548">✓</td></tr> <tr> <td data-bbox="295 548 1109 609">Has less complex circuitry</td><td data-bbox="1109 548 1225 609">✓</td><td data-bbox="1225 548 1342 609"></td></tr> <tr> <td data-bbox="295 609 1109 743">Requires higher power consumption under low levels of access, which is significant when used in battery-powered devices</td><td data-bbox="1109 609 1225 743">✓</td><td data-bbox="1225 609 1342 743"></td></tr> <tr> <td data-bbox="295 743 1109 837">Requires data to be refreshed occasionally so it retains the data</td><td data-bbox="1109 743 1225 837">✓</td><td data-bbox="1225 743 1342 837"></td></tr> </tbody> </table>	Statement	DRAM	SRAM	Does not need to be refreshed as the circuit holds the data while the power supply is on		✓	Mainly used in cache memory of processors where speed is important		✓	Has less complex circuitry	✓		Requires higher power consumption under low levels of access, which is significant when used in battery-powered devices	✓		Requires data to be refreshed occasionally so it retains the data	✓		5
Statement	DRAM	SRAM																		
Does not need to be refreshed as the circuit holds the data while the power supply is on		✓																		
Mainly used in cache memory of processors where speed is important		✓																		
Has less complex circuitry	✓																			
Requires higher power consumption under low levels of access, which is significant when used in battery-powered devices	✓																			
Requires data to be refreshed occasionally so it retains the data	✓																			

9608_w16_qp_13_sc_2.a

QUESTION

2 (a) State **two** differences between Static RAM (SRAM) and Dynamic RAM (DRAM).

1

.....

2

.....[2]

MARK SCHEME

2 (a) Any **two** from:

- DRAM has to be refreshed / charged
// SRAM does not request a refresh
- DRAM uses a single transistor and capacitor
// SRAM uses more than one transistor to form a memory cell
// SRAM has more complex circuitry
- DRAM stores each bit as a charge
// SRAM each bit is stored using a flip-flop / latch
- DRAM uses higher power(because it requires more circuitry for refreshing)
// SRAM uses less power (no need to refresh)
- DRAM less expensive (to purchase / requires fewer transistors)
// SRAM is more expensive (to buy as it requires more transistors)
- DRAM has slower access time / speed (because it needs to be refreshed)
// SRAM has faster access times
- DRAM can have higher storage / bit / data density
// SRAM has lower storage / bit / data density
- DRAM used in main memory
// SRAM used in cache memory

[2]

9608_w16_qp_12_sc_6.b

QUESTION

(b) State **three** differences between Dynamic RAM (DRAM) and Static RAM (SRAM).

- 1
-
- 2
-
- 3
-[3]

MARK SCHEME

(b) **THREE** from:

- DRAM has to be refreshed / charged
// SRAM does not request a refresh
- DRAM uses a single transistor and capacitor
// SRAM uses more than one transistor to form a memory cell
// SRAM has more complex circuitry
- DRAM stores each bit as a charge
// SRAM each bit is stored using a flip-flop/latch
- DRAM uses higher power (because it requires more circuitry for refreshing)
//SRAM uses less power (no need to refresh)
- DRAM less expensive (to purchase / requires fewer transistors)
// SRAM is more expensive (to buy as it requires more transistors)
- DRAM has slower access time / speed (because it needs to be refreshed)
// SRAM has faster access times
- DRAM can have higher storage / bit / data density
// SRAM has lower storage / bit / data density
- DRAM used in main memory
// SRAM used in cache memory

[3]

9608_w16_qp_11_sc_2.a

QUESTION

2 (a) State **two** differences between Static RAM (SRAM) and Dynamic RAM (DRAM).

- 1
-
- 2
-[2]

MARK SCHEME

2 (a) Any **two** from:

- DRAM has to be refreshed / charged
// SRAM does not request a refresh
- DRAM uses a single transistor and capacitor
// SRAM uses more than one transistor to form a memory cell
// SRAM has more complex circuitry
- DRAM stores each bit as a charge
// SRAM each bit is stored using a flip-flop / latch
- DRAM uses higher power(because it requires more circuitry for refreshing)
// SRAM uses less power (no need to refresh)
- DRAM less expensive (to purchase / requires fewer transistors)
// SRAM is more expensive (to buy as it requires more transistors)
- DRAM has slower access time / speed (because it needs to be refreshed)
// SRAM has faster access times
- DRAM can have higher storage / bit / data density
// SRAM has lower storage / bit / data density
- DRAM used in main memory
// SRAM used in cache memory

[2]

QUESTION

- 4 (a) There are two types of RAM: dynamic RAM (DRAM) and static RAM (SRAM).

Five statements about DRAM and SRAM are shown below.

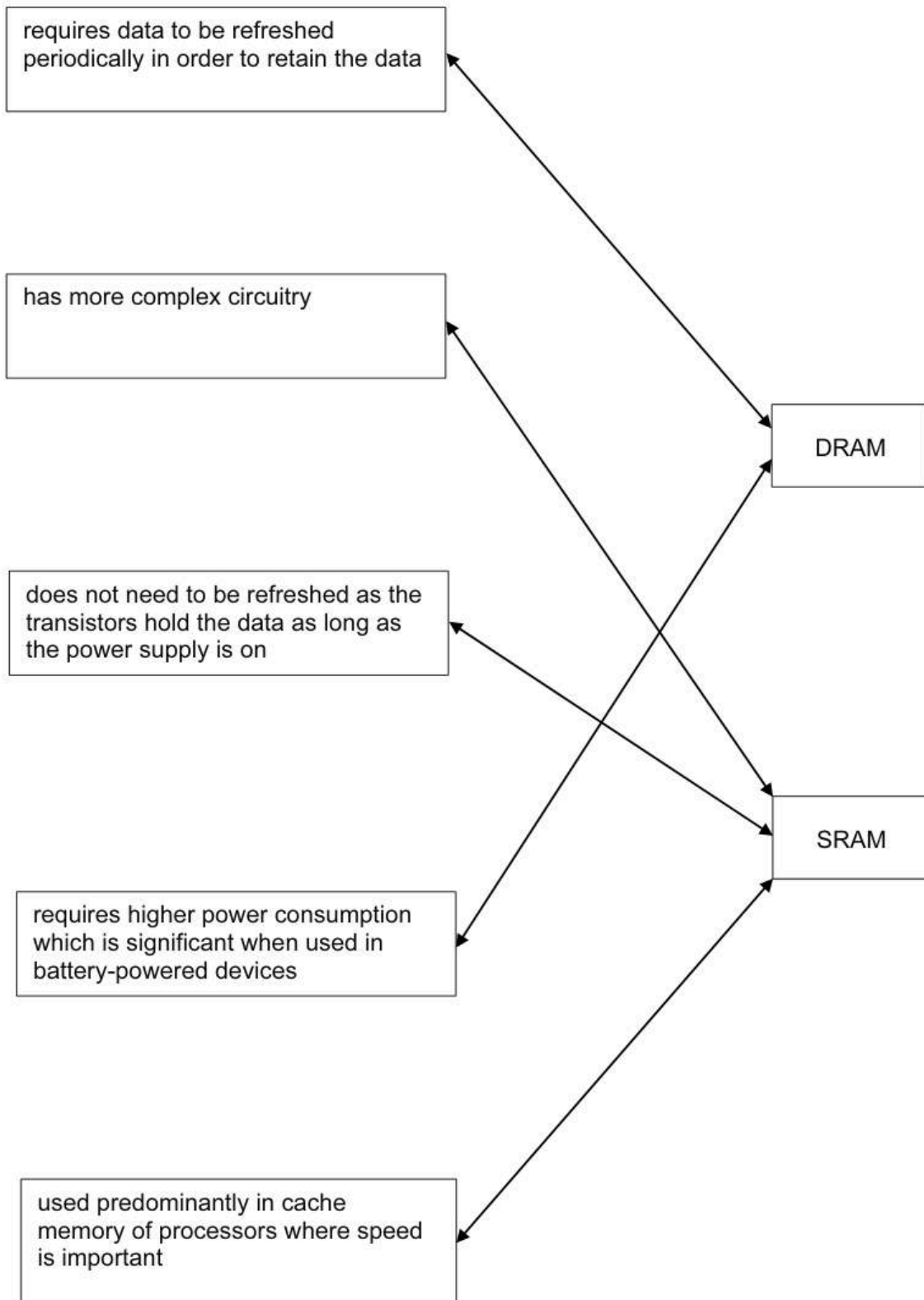
Draw a line to link each statement to the appropriate type of RAM.

Statement	Type of RAM
requires data to be refreshed periodically in order to retain the data	
has more complex circuitry	DRAM
does not need to be refreshed as the circuit holds the data as long as the power supply is on	
requires higher power consumption which is significant when used in battery-powered devices	SRAM
used predominantly in cache memory of processors where speed is important	

[5]

MARK SCHEME

4 (a)



[5]

Explain the difference between Programmable ROM (PROM), Erasable Programmable ROM (EPROM) and Electrically Erasable Programmable ROM (EEPROM)

Show an understanding of monitoring and control systems

9608_w15_qp_13_sc_5.i

QUESTION

- 5 A computer system in a control room is used to monitor earthquake activity.

An earthquake zone has a number of sensors to detect seismic activity.

The system detects when seismic activity is greater than 3 on the Richter Scale. Whenever this happens, a printer in the control room prints a report.

- (i) Identify the steps that are required in this monitoring system.

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 5 (i) any **four** from:

- sensors send signals / data to a computer
- data / signal converted to digital (using an ADC)
- computer calculates the activity value based on sensor signal strength / data
- if calculated value > 3,
- determine location of sensor
- build up a map of seismic activity
- location and magnitude sent to printer
- ... via USB port / wireless link
- monitoring is continuous

[4]

9608_w15_qp_13_sc_5.ii

QUESTION

- (ii) When the system detects high activity, operators may need to respond rapidly. A printer is useful for hard copies, but may not be the best way to inform operators.

Give a reason why.

.....
.....[1]

MARK SCHEME

- (ii) – output / hard copy not picked up in good time [1]

9608_w15_qp_13_sc_5.iii

QUESTION

- (iii) Name an alternative output device for this monitoring system and give a reason for your choice.

Output device

Reason for choice

.....
.....[2]

MARK SCHEME

- (iii) 1 mark for name + 1 mark for reason for choice

- speaker / buzzer / alarm bell
- gives control room operators an audible warning
- flashing lights
- gives control room operators an visual warning
- monitor
- use of red / flashing colours on monitor gets the attention of operators [2]

QUESTION

- 5 A computer system in a control room is used to monitor earthquake activity.

An earthquake zone has a number of sensors to detect seismic activity.

The system detects when seismic activity is greater than 3 on the Richter Scale. Whenever this happens, a printer in the control room prints a report.

- (i) Identify the steps that are required in this monitoring system.

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 5 (i) any **four** from:

- sensors send signals / data to a computer
- data / signal converted to digital (using an ADC)
- computer calculates the activity value based on sensor signal strength / data
- if calculated value > 3,
- determine location of sensor
- build up a map of seismic activity
- location and magnitude sent to printer
- ... via USB port / wireless link
- monitoring is continuous

[4]

QUESTION

- (ii) When the system detects high activity, operators may need to respond rapidly. A printer is useful for hard copies, but may not be the best way to inform operators.

Give a reason why.

.....

.....[1]

MARK SCHEME

(ii) – output / hard copy not picked up in good time

[1]

9608_w15_qp_11_sc_5.iii

QUESTION

(iii) Name an alternative output device for this monitoring system and give a reason for your choice.

Output device

Reason for choice

.....

.....

[2]

MARK SCHEME

(iii) 1 mark for name + 1 mark for reason for choice

- speaker / buzzer / alarm bell
- gives control room operators an audible warning
- flashing lights
- gives control room operators an visual warning
- monitor
- use of red / flashing colours on monitor gets the attention of operators

[2]

9608_s15_qp_12_sc_4.a.i

QUESTION

4 (a) Sensors are one type of input device.

For each of the following situations, name a **different** sensor that could be used.

(i) air conditioning in an office building

.....[1]

MARK SCHEME

(i) temperature/thermistor

[1]

9608_s15_qp_12_sc_4.a.ii

QUESTION

(ii) maintaining correct growing conditions in a greenhouse

.....[1]

MARK SCHEME

(ii) moisture, humidity, light/photodiode, temperature, pH

[1]

9608_s15_qp_12_sc_4.a.iii

QUESTION

(iii) detecting an intruder in a building

.....[1]

MARK SCHEME

(iii) sound/acoustic, infrared, pressure, motion, microwave

[1]

9608_s15_qp_11_sc_4.a.i

QUESTION

4 (a) Sensors are one type of input device.

For each of the following situations, name a **different** sensor that could be used.

(i) air conditioning in an office building

.....[1]

MARK SCHEME

(i) temperature/thermistor

[1]

9608_s15_qp_11_sc_4.a.ii

QUESTION

(ii) maintaining correct growing conditions in a greenhouse

.....[1]

MARK SCHEME

(ii) moisture, humidity, light/photodiode, temperature, pH

[1]

9608_s15_qp_11_sc_4.a.iii

QUESTION

(iii) detecting an intruder in a building

.....[1]

MARK SCHEME

(iii) sound/acoustic, infrared, pressure, motion, microwave

[1]